SPRINT 0

why are we here?

Project Name: Sudo!
Team: The Avengers

Team Members

- Dmytro Senyk – Product Owner
- Bogdan Tabalae – DevOps
- Kostiantyn Panasenko - Quality Tester
- Tomasz Touma – Scrum Master

PROBLEM & IDEA

Problem

Many Sudoku apps use fixed difficulty levels, which do not adapt to a player's skill level. This can lead to boredom for experienced players or frustration for beginner

Our Idea

Create a Smart Adaptive sudoku app that adjusts puzzle difficulty based on user performance and progress.

ELEVATOR PITCH

Sudo is a personalizd brain training Sudoku app.

The system analyses:

- Solving Time
- Mistakes
- Hint usage

Based on this data, the app dynamically adjusts difficulty to provide the right level of challenge, helping users improve logical thinking while staying engaged

KEY FEATURES & BENEFITS

KEY FEATURES

- ADAPTIVE DIFFICULTY SYSTEM
- SMART HINTS AND MISTAKE DETECTION
- PROGRESS TRACKING DASHBOARD
- DAILY CHALLENGES

BENEFITS

- PERSONALISED LEARNING EXPERIENCE
- IMPROVED LOGICAL REASONING
- INCREASED USER ENGAGEMENT

SCOPE

- In scope
  - Core Sudoku gameplay
  - Adaptive difficulty
  - Hint System
  - User profiles
  - Session management
- Out of Scope
  - Multiplayer modes
  - Social features
  - Cloud Sync

SCRUM TEAM ROLES

- Product Owner
  - Defines product vision
  - Manages and prioritizes backlog
- Scrum Master
  - Facilitates scrum process
  - Removes blockers
- DevOps
  - Manages GitHub Repo
  - Prepares CI/CD pipeline
- Quality Tester
  - Defines testing strategy
  - Ensures software quality

RELEASE PLAN

Release plan can help us define:

Sprint length

Team velocity

Number of sprints

Target release date

Plan Sprint length: 1 week

Sprint 0 for preparation

5 development sprints

Target release: **End of Sprint 5**

SPRINT 0 ACTIVITIES

Sprint 0 focused on project preparation:

- Setup GitHub repository
- Configure Jira project
- Prepare development enviroment
- Refine product backlog
- Define Definition on Ready & Definition of Done

This ensured the team was ready to start development

PRODUCT ROADMAP

Development was divided into phases:

Phase 1

Core gameplay

Phase 2

User experience improvement

Phase 3

Engagement features (tracking & challenges)

Phase 4

Testing and optimistaion

SYSTEM ARCHITECTURE

The application follows a client-based architecture

Main components:

User inteface

- Sudoku grid
- Controls and menus

Game logic engine

- Puzzle generation
- Difficulty management
- Validation logic

Data layer

- Local storage for progress

TOOLS & PLANNING

Tools used:

- **Jira**

Used for product backlog and sprint planning

- **GitHub**

Source code repository

- **Microsoft Project**

Gantt chart and timeline

We also created:

- Release burndown chart
- Sprint backlog
- Roadmap planning

RISKS & CONCLUSION

Key Risks:

- Compexity of Sudoku generator
- Difficulty balancing adaptive system
- Team coordination
- React + firebase integration

Mitigation

- Weekly Scrum meetings
- Early testing
- Focus on core features first

Conclusion:

Sprint 0 allowed our team to establish planning, tools, architecture, and Scrum structure before starting development.